

Unit 4 Question Bank

MCQ Questions (1 Mark each question)

1. A Packet Filtering Firewall operates rapidly by checking only the _____ of a packet.

- A) Application layer
- B) Data
- C) **Header**
- D) Content

2. The type of VPN that allows employees to securely connect to their company's network from remote locations is called _____.

- A) Cloud VPN
- B) Site-to-Site VPN
- C) **Remote Access VPN**
- D) Mobile VPN

3. One major limitation of using a Virtual Private Network (VPN) is that _____.

- A) VPNs always increase internet speed
- B) VPN usage is banned in all countries
- C) **Premium VPN services can be expensive**
- D) VPNs do not provide any security for data transmission

4. In a conventional encryption scheme, which of the following is NOT one of the five ingredients?

- A) Plain Text
- B) Encryption Algorithm
- C) **Public Key**
- D) Decryption Algorithm

5. In Public Key Cryptography, the purpose of the public key is to _____.

- A) It is used for decryption by the recipient.
- B) **It is used for encryption by the sender.**
- C) It is kept private by the owner.
- D) It is used to verify the digital signature.

6. A primary use case for a Digital Signature in Public Key Cryptography is to _____.

- A) Encrypt a message with the recipient's private key
- B) **Ensure data integrity and authenticate the sender**
- C) Exchange session keys securely
- D) Encode the message using a cipher text

7. A Substitution Cipher _____.

- A) Changes the order of the plaintext without altering the symbols.
- B) **Replaces each letter or group of letters in the plaintext with another letter or group**

of letters.

- C) Uses mathematical algorithms to disguise the plaintext.
- D) Only rearranges the positions of letters in the message.

8. The process of converting plaintext into an encrypted message is known as _____.

- A) Decryption
- B) Cryptanalysis
- C) **Encryption**
- D) Compression

9. In Public Key Cryptography, the private key is _____.

- A) Made public for anyone to use
- B) **Known only to the owner and used for decryption**
- C) Shared openly with all participants
- D) Used to encrypt messages by the sender

10. A Transposition Cipher differs from a Substitution Cipher in that it _____.

- A) **Changes the order of the plaintext symbols**
- B) Replaces each letter with a corresponding symbol
- C) Uses a key to transform the plaintext into cipher text
- D) Performs mathematical operations on the message

11. The primary role of a cryptanalyst in the context of encryption is to _____.

- A) Encrypt messages using secret keys
- B) **Analyze cipher text and restore the original message**
- C) Distribute public keys
- D) Generate secret keys

12. Which of the following is a characteristic of Network-Based IDS?

- a) Can monitor an entire network with minimal resources
- b) Can analyze encrypted data easily
- c) **Requires significant host resources**
- d) Monitors individual hosts for signs of intrusion

13. Which of the following is a primary function of the Physical Layer (Layer 1) in the OSI model?

- a) Handling error detection and correction
- b) **Transmitting raw bits over a physical medium**
- c) Routing data across networks
- d) Ensuring reliable end-to-end communication

14. What is the main role of the Data Link Layer (Layer 2)?

- a) Encrypting data during transmission
- b) Establishing and maintaining links between directly connected nodes
- c) **Handling IP addressing and routing**
- d) Ensuring data integrity and security at the application layer

15. Which of the following wireless security protocols provides the strongest encryption and protection against brute force attacks?

- a) WEP
- b) WPA
- c) WPA3**
- d) WPA2

16. What is one of the most common ways to protect a Wi-Fi network from unauthorized access?

- a) Using only default passwords
- b) Disabling encryption
- c) Using WPA2 or WPA3 encryption protocols**
- d) Connecting devices to unsecured networks

Short question (2 Marks each)

1. What is firewall?
2. List the types of VPN.
3. List the benefits of VPN.
4. Explain Substitution Ciphers.
5. Explain Transposition Ciphers.
6. List the characteristics of VPN.
7. What is cryptanalysis?
8. List the public key encryption scheme ingredients.

Long Questions (6 Marks each)

1. Explain internet Architecture in detail.
2. What is wireless security? Explain types of wireless security.
3. Explain E-commerce security needs.
4. What is VPN? Explain its Types, Benefits, limitations and how it works.
5. Explain firewall in detail.